Adaptations for extremes

Schimper made a close study of plant physiology—the structures of plants and how they had adapted to varying temperature and moisture conditions. He was particularly interested in plants growing in extreme climatic conditions. Salty environments, for example, require plants to survive high levels of soil and water salinity. Schimper found that vegetation growing on the coastal mangroves of Brazil, on Caribbean and Sri Lankan beaches, and in sulfur-emitting volcanic craters in Java, were similarly tolerant to salt.

Schimper also studied how plants coped in the challenging conditions of arid environments. He found that plants growing in hot, dry places had evolved “varied contrivances for regulating the passage of water.” To illustrate this, he chose a type of vegetation with tough leaves, short internodes (the distances between the leaves along a stem), and leaf orientation parallel or oblique to direct sunlight. This type grew in various parts of the world, where arid conditions meant that water was scarce. The name Schimper gave to these plants—sclerophyll, from the Greek words *skleros* (“hard”) and *phullon* (“leaf”)—is still used today.

Epiphytes, plants that grow on the surface of other plants and derive their moisture and nutrients from the air or rain, also fascinated Schimper. He observed epiphytes such as Spanish moss growing in the southern US and the Caribbean islands and similar species in South America, South Asia, and southeast Asia. He found that they were linked by warm temperatures and year-round moisture—traits of what he called a tropical rain forest.

Although the broad geographic divisions devised by Schimper still hold true, there is now a better understanding of how vegetation develops in response to many different stimuli beyond simple climatic differences. For example, measures of potential water evaporation into the atmosphere, water surplus, and water deficit, which can be combined in a moisture